

Full-Period Concentration in Mersenne Rule 90

Route-A structural certificate C155

Abstract

For Rule 90 on Mersenne rings $L = 2^r - 1$, the periodic image contains 2^{L-1} states and every period divides L . We prove that exact period L has probability tending to one, the total cycle count is asymptotic to $2^{L-1}/L$, and the cycle-averaged period is asymptotic to L . The neighboring power-of-two family remains nilpotent. These are all-size theorems, not extrapolations from the finite replay ledger.

Keywords: cellular automaton; Rule 90; Mersenne circumference; exact period; fixed subspace; Burnside lemma; orbit concentration.

中文摘要

本文研究长度 $L = 2^r - 1$ 的环上 Rule 90。其周期集是维数 $L - 1$ 的线性像，且所有轨道周期整除 L 。我们证明任意非恒等时刻的固定子空间只依赖 $d = \gcd(j, L)$ ，其维数不超过 $2d \leq 2L/3$ 。由此得到：在周期像上均匀取点时，周期小于 L 的概率至多为 $2L2^{-L/3}$ ，因而精确满周期的概率趋于 1。Burnside 引理进一步给出总原始轨道数渐近于 $2^{L-1}/L$ ，并说明按轨道等权计算的平均周期除以 L 后趋于 1。作为匹配对照，二次幂长度族仍为幂零系统。这些结论是 Mersenne 尺度上的精确有限体积定理，不是目标行列式匹配。
关键词：元胞自动机；Rule 90；Mersenne 周长；精确周期；固定子空间；Burnside 引理；轨道集中。

1 Periodic image and proper fixed spaces

On $R_L = \mathbb{F}_2[x, x^{-1}]/(x^L - 1)$, Rule 90 is multiplication by $a = x + x^{-1}$. Frobenius gives $a^{L+1} = a$. Since $x^L + 1$ is squarefree for odd L and $\gcd(x^L + 1, (x + 1)^2) = x + 1$, multiplication by a has a one-dimensional kernel. Thus the periodic set is $V = \text{im } a$, with $|V| = 2^{L-1}$, and $g = a|_V$ satisfies $g^L = I$. Conversely, a positive-time fixed state already belongs to V , so every realized period divides L .

For $1 \leq j < L$, put $d = \gcd(j, L)$. The identity $\gcd(U^j - 1, U^L - 1) = U^d - 1$ and Bézout give, after substitution $U = g$ and use of $g^L = I$,

$$\ker(g^j - I) = \ker(g^d - I).$$

Clearing the Laurent multiplier for time d gives $(x^2 + 1)^d + x^d$, of degree $2d$. Because a proper divisor of odd L is at most $L/3$,

$$\dim \ker(g^j - I) \leq 2d \leq \frac{2L}{3}. \quad (1)$$

2 Concentration and Burnside

Every state of period less than L belongs to one of the proper-time fixed spaces because every realized period divides L . Hence

$$\Pr_V(\text{per} < L) \leq \frac{(L-1)2^{2L/3}}{2^{L-1}} \leq 2L2^{-L/3} \rightarrow 0. \quad (2)$$

If C_L is the total number of primitive cycles, Burnside gives

$$C_L = \frac{1}{L} \sum_{j=0}^{L-1} \# \text{Fix}_V(g^j), \quad 0 \leq \frac{LC_L}{2^{L-1}} - 1 \leq 2L2^{-L/3}. \quad (3)$$

The cycles partition V , so their mean length when each cycle is weighted once is $|V|/C_L = 2^{L-1}/C_L$. After division by L , this is the reciprocal of $LC_L/2^{L-1}$ and tends to one.

The uniform estimate can be trivial at the smallest size: for $L = 3$ the restriction is the identity and there is no exact period-three state. This finite exception is retained in the ledger and does not affect the limit. Exact polynomial-gcd and Möbius rows through $r = 8$ contain 26 divisor-period and 494 proper-time cells. The independent checker passes 2,291 assertions, SymPy passes 2,255 checks, replay is byte-identical, and 54 hostile cases are rejected. For the matched control $L = 2^s$, $a^{L/2} = 0$, so only zero is periodic.

The strict tuple is (A1_WEAK, A2_FAIL, A3_FAIL, A4_FAIL). We claim no thermodynamic determinant, target divisor, target functional equation or counting law, arithmetic/local factor, root number, automorphy, natural operator lift, Hilbert–Pólya operator, or Route-B authorization. Scope: NO_BAD_EULER_OR_ROOT_NUMBER.

Declarations

Data and code availability. The package-local `results/c155_rule90_concentration_evidence.json` and `code/` directory contain the exact certificate, deterministic producer, independent checker, symbolic cross-check, replay, and mutation audit.

Ethics. This mathematical and symbolic-computational study uses no human participants, animals, personal data, clinical records, or sensitive data; research ethics approval is not applicable.

Author contributions (CRediT). This anonymous technical certificate does not assign individual authorship. Package provenance records Conceptualization, Formal analysis, Software, Validation, Writing—original draft, and Writing—review and editing at the artifact level. AI systems are not authors.

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AI-use disclosure. An AI coding assistant supported drafting, exact-code development, and internal proof checking. It was not an external reviewer. Released claims are exposed to deterministic reproduction, independent checking, symbolic cross-checking, and hostile mutation tests in the package.